ENERGY CONSERVATION

Smart Community Energy Portal

Project on energy conservation
Social Internship (BE05000011) | B.E. Sem V, Computer Engineering

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SSASIT, Surat
Gujarat Technological University
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Introduction

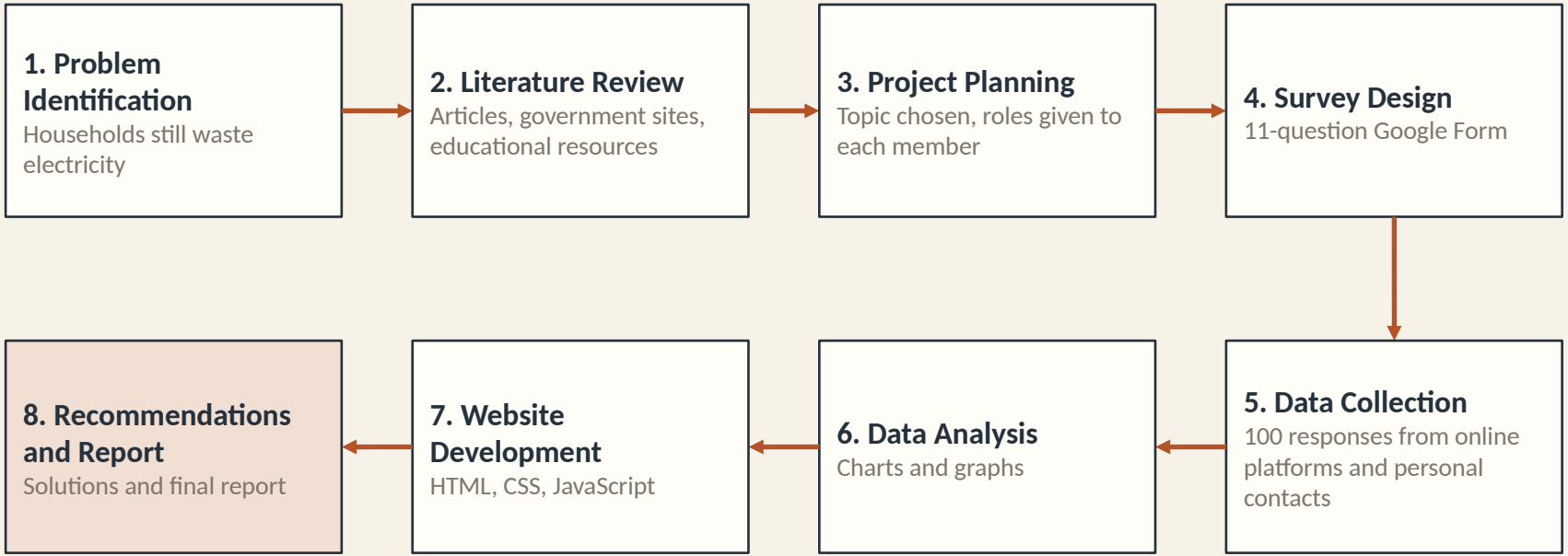
Why this project?

- Electricity use is growing every year in homes, schools and industries.
- Small habits like switching off lights and using LED bulbs save money and reduce pollution.
- Many people know this, but do not follow it daily.
- So we made a website and ran a community survey.

Our objectives

- Build an easy web portal on energy saving
- Spread awareness about using electricity wisely
- Run a survey using Google Forms
- Study the answers with charts
- Suggest simple ways to reduce waste

Methodology



Total duration: 15-day internship (15 June to 30 June 2026)

Project Timeline

Phase	Activity	Deliverable	Duration
Planning	Team formation, scope definition, tool setup	Project plan, role assignment	Day 1-2
Research	Literature review, content development, survey design	Research notes, website content, questionnaire	Day 2-4
Development	Website coding, testing, integration	Functional multi-page website	Day 3-9
Survey	Survey distribution, response monitoring	100 survey responses	Day 5-12
Analysis & Reporting	Data analysis, visualization, report writing	Report, presentation, findings	Day 10-15

Community Survey at a Glance

11	100	~7 days	Anonymous
Questions	Responses	19-27 June 2026	Response format

How we shared the survey

- WhatsApp groups
- Social media (Instagram and Facebook)
- College networks
- QR code on the portal website

Survey details

- Made on Google Forms
- Question types: multiple choice, multi-select, rating scale (1-5)
- Checked with Google Forms analytics and manual analysis

Who Answered the Survey

Area and people

- Done in and around Surat, Gujarat
- Reached through SSASIT college network, WhatsApp groups and social media
- Students, working professionals, homemakers and senior citizens answered
- So the results show many groups, not only one

What we observed

- People were happy to take part once we explained the purpose
- Students replied fastest on WhatsApp and social media
- Homemakers and senior citizens were reached by personal contact and the QR code
- Using many channels worked better than one

Survey QR and Responses



Scan to fill the survey

Google Form responses

All 100 answers collected from the community are in this link.

Open the responses

https://docs.google.com/forms/d/1_TDC_y_KtZnl_tVolGjlV7auP5n59YlOnN5wQT6v9KQ/edit#responses

Key Findings from 100 Responses

84%

agree that saving energy reduces pollution

71%

use natural light always or often

64%

use LED bulbs at home

56%

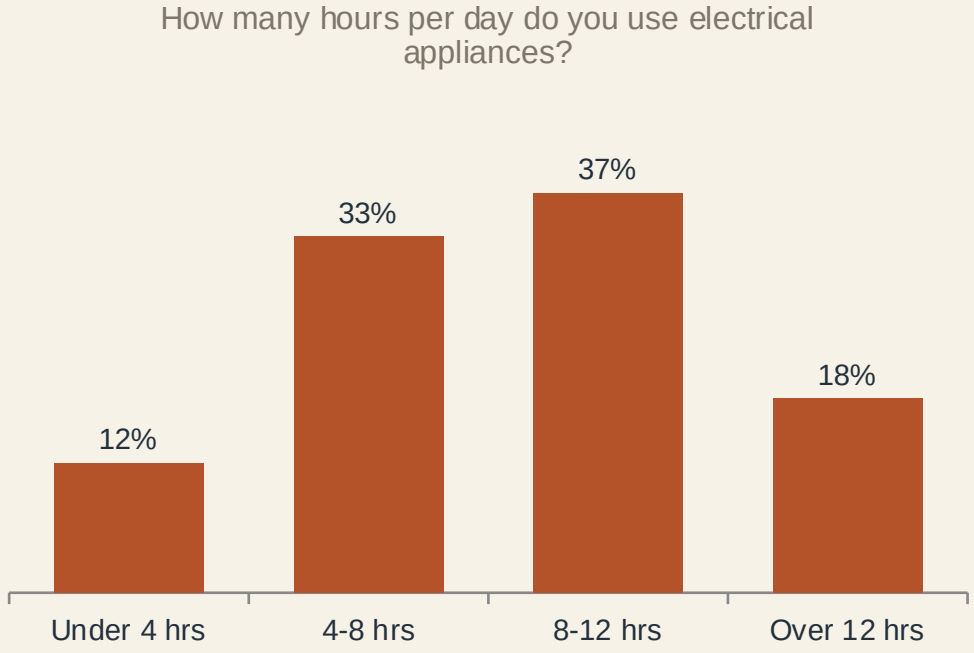
noticed a rise in their electricity bill

48%

say too much AC and fan use is the top cause of wastage

Source: Community Energy Conservation Awareness Survey, 100 respondents, June 2026

Daily Appliance Usage

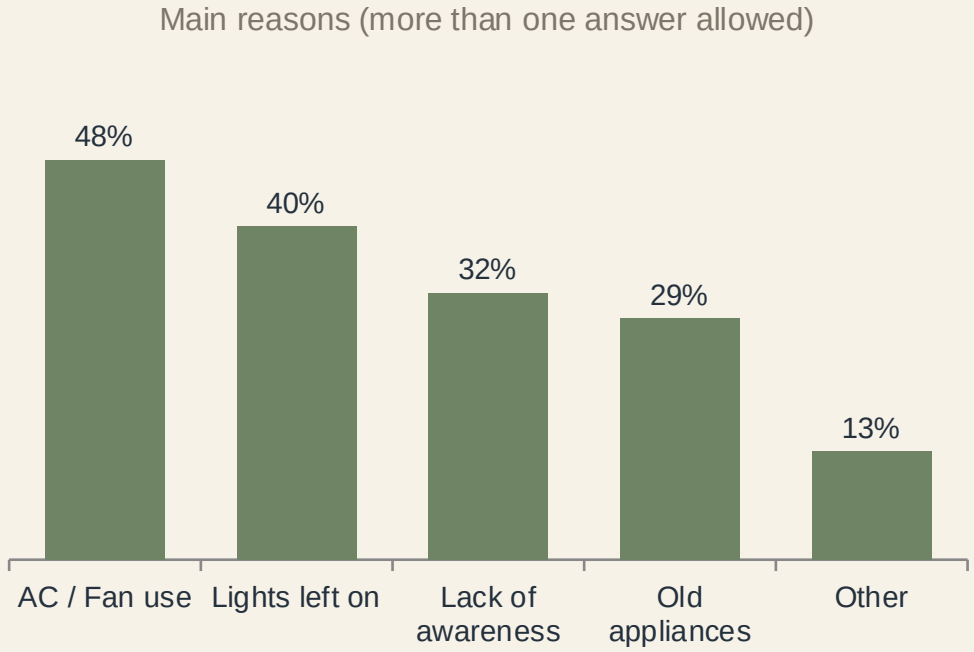


55%

use appliances 8 or more hours a day

- 18% use them for more than 12 hours, so their bills are at higher risk
- Only 12% use them under 4 hours
- Better appliances and planned use can cut consumption

Main Reasons for Energy Wastage



- Excess AC and fan use (48%) is the top reason: Surat has hot summers
- Leaving lights on (40%) is a habit that awareness can change
- Lack of awareness (32%) supports the purpose of this project
- Old appliances (29%) need to be upgraded
- So cooling habits are the first priority

Root Causes of Energy Wastage

1	**Excessive cooling use**	AC and fans at very low settings or in empty rooms (48%)
2	**Behavioural waste**	Lights left on (40%); 52% sometimes or never unplug devices
3	**Incomplete LED transition**	36% still use tube lights, CFLs or incandescent bulbs
4	**Awareness program gap**	57% have never joined an awareness program
5	**Awareness-behaviour gap**	63% rate themselves 4-5 in awareness, yet habits stay poor

SWOT Analysis

Strengths

- High awareness of the pollution link (84%)
- Strong LED adoption (64%)
- Good natural-light use (71%)
- College-led credibility

Weaknesses

- 55% use appliances 8+ hrs a day
- Low unplugging habit in 52% of homes
- 57% never joined an awareness program

Opportunities

- 70% want government and schools to lead
- Smart plugs and LED-tube retrofits give quick wins
- Digital platforms can reach the 57%

Threats

- Rising bills without change in behaviour
- Continued use of old appliances
- Cooling demand likely to keep growing

Our Prototype: The Portal

What the portal offers

- Energy-conservation information
- Practical energy-saving tips
- Renewable vs non-renewable energy education
- Embedded community survey with QR access
- Simple, responsive design for mobile and computer

Tools used

Tool	Purpose
HTML	Structure of the website
CSS	Design and styling
JavaScript	Interactive features
Google Forms	Community survey
Google Sheets	Data collection and analysis
Microsoft Excel	Charts and graphs
Microsoft Word	Report preparation

Solutions: Cooling and Behaviour

Excessive Cooling Energy Use

- Set AC at 24-26 C and use a fan so the thermostat can be 2-3 degrees higher
- Turn off AC and fans when leaving a room
- Use curtains, blinds or window films to cut heat
- Replace ACs older than 5-7 years with 5-star inverter units (30-40% less electricity)

Behavioural Energy Waste

- "Turn Off All" reminders near switches and a "last one out" rule
- Smart power strips and motion-sensor lights
- Weekly or monthly energy-tracking challenges
- School programs that teach energy-saving habits

Solutions: LED and Awareness

Incomplete LED Transition

- Tell tube-light users about LED tubes (30-50% savings)
- Bulk buying of LED products at lower prices
- Promote the UJALA scheme
- Safe disposal of old tube lights and CFLs, which contain mercury

Awareness Program Gap

- Reliable, central information on the portal
- Encourage people to join community surveys
- Share practical energy-saving tips
- Educational content and interactive pages with visual learning

Impact Assessment

Social

- 100 people directly took part in the survey
- Students, professionals, homemakers and senior citizens joined
- Started talks across generations about using electricity wisely
- 70% say institutions and government should lead

Economic

- 56% reported higher electricity bills
- Better AC settings, appliance upgrades and less standby power can lower bills
- 19% say lower bills is their main motivation

Environmental

- Less electricity use lowers demand on fossil-fuel power
- Fewer greenhouse gases and less air pollution
- LED use, less cooling waste and low standby power reduce the carbon footprint

Advantages and Limitations

Feasibility

- Uses low-cost tools: LED bulbs, smart plugs, awareness content
- No special hardware needed
- Free tools like Google Forms and Sheets, so no big budget

Advantages

- Targets the exact causes found in the survey
- Evidence-based, not generic
- Matches the wish for government and school-led action

Limitations

- Only 100 responses in a short time, may not represent all of Surat or Gujarat
- No hardware or real-time energy monitoring
- Effect on actual electricity use not measured

Conclusion and Future Scope

Conclusion

Seven Computer Engineering students completed the Smart Community Energy Portal in the 15-day internship (15 to 30 June 2026). The 100 survey responses show that people value energy conservation and use LED lighting and daylight, but heavy AC and fan use, daily habits and low program participation remain challenges. The portal helps by spreading awareness through a digital platform.

Future scope

- Mobile application
- Energy consumption calculator
- AI-based personalised saving tips
- Support for English, Hindi and Gujarati
- More content, quizzes and updates

Thank you